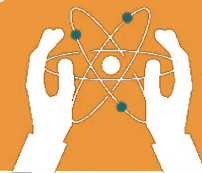


UNIT 4

FUNDAMENTALS OF CHEMISTRY

Learning objectives:

- Understand chemistry's role in cooking, cleaning, and medicine.
- Learn about the contributions of Aristotle, Jabir ibn Hayyan, Boyle, and Curie.
- Identify Physical, Organic, Inorganic, Biochemistry, and Industrial Chemistry.



4.1 Introduction

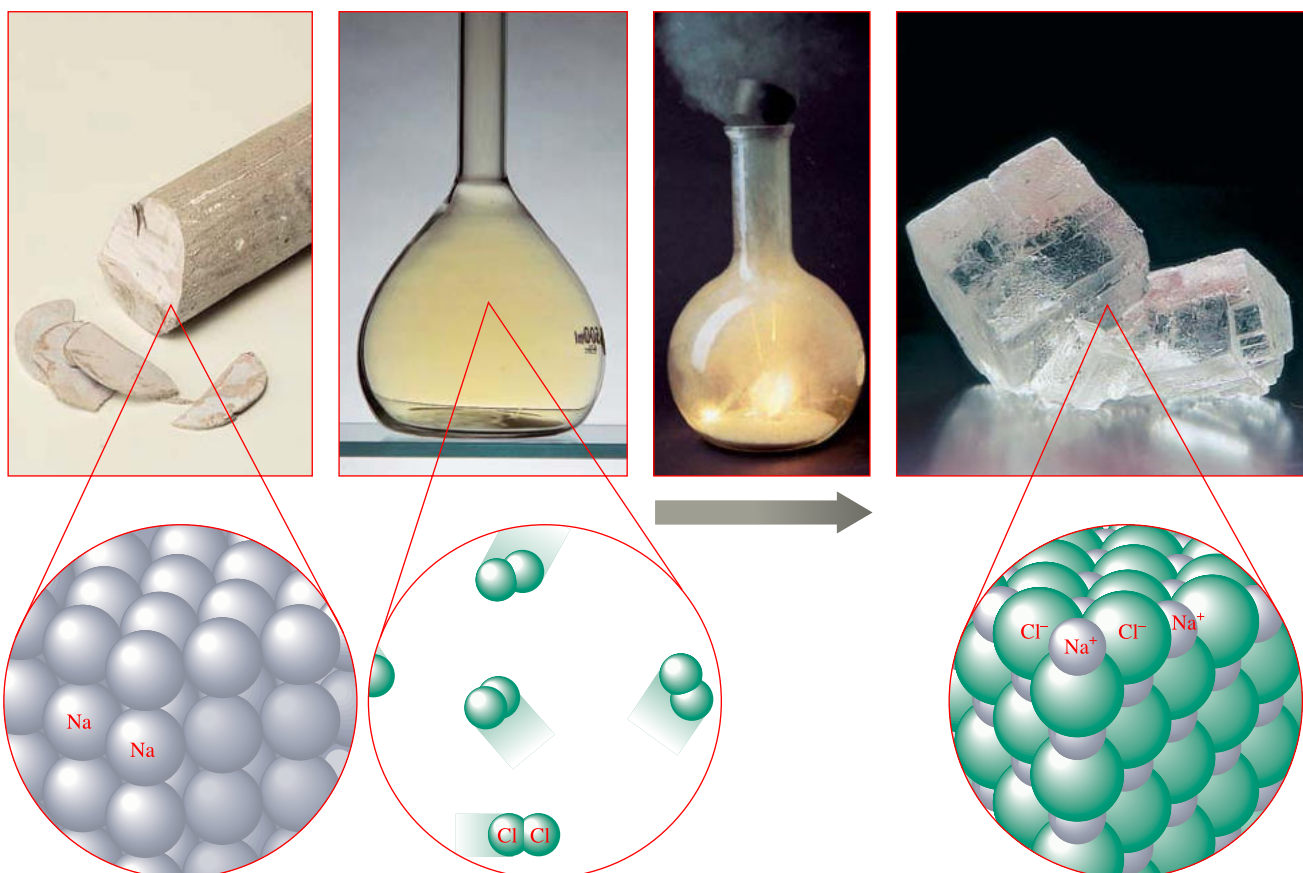
Definition of Chemistry

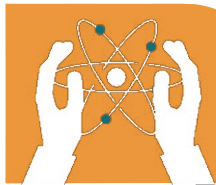
Chemistry is the branch of science that studies matter, its properties, changes, and the reactions when substances combine or break apart. It helps us understand how substances interact and how new ones are formed.

Importance of Chemistry in Daily Life

Chemistry plays a major role in everyday life. It helps us understand how things around us work, from the food we eat to the medicines we take.

For example, chemistry helps us cook, clean, and even make our homes safe by understanding the properties of materials.





4.2 Historical Background of Chemistry

Contributions of Early Scientists

- **Aristotle and Plato:** These ancient Greek philosophers believed that matter was made up of four basic elements: earth, water, air, and fire.



- **Jabir ibn Hayyan:** A famous Muslim scholar who made major contributions to alchemy (the precursor to chemistry). He is known for his work in the development of laboratory techniques, such as distillation.



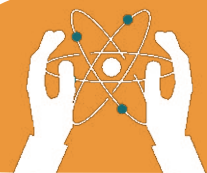
Evolution of Chemistry through Modern Scientists

- **Robert Boyle:** Known as the “father of modern chemistry,” Boyle proposed that the volume of a gas is inversely related to its pressure (Boyle's Law).



- **Marie Curie:** A famous scientist who discovered radium and polonium, contributing to the field of radioactivity.





4.3 Branches of Chemistry

1. Physical Chemistry:

Physical chemistry studies how matter behaves at the molecular and atomic levels, focusing on properties like temperature, pressure, and volume during chemical reactions.

2. Organic Chemistry:

Organic chemistry focuses on carbon-containing compounds, particularly those found in living organisms, like proteins and carbohydrates.

3. Inorganic Chemistry:

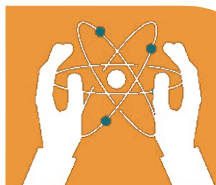
Inorganic chemistry deals with substances that do not contain carbon, such as metals, minerals, and salts.

4. Biochemistry:

Biochemistry explores the chemical processes within living organisms, including how enzymes function and how cells produce energy.

5. Industrial Chemistry:

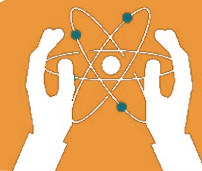
Industrial chemistry involves large-scale chemical production, including processes for making plastics, fertilizers, and medicines.



EXERCISES:

Fill in the blanks.

1. _____ is the branch of science that studies matter and its changes.
2. The early Greek philosophers, _____ and _____, believed that matter was made up of four elements.
3. Robert Boyle is known for _____'s law, which deals with the relationship between the pressure and volume of gases.
4. _____ chemistry is concerned with carbon-containing compounds.
5. _____ chemistry helps us understand the chemical processes in living organisms.
6. The process of _____ involves studying the effect of temperature on chemical reactions.
7. _____ chemistry studies the behaviour of matter on a molecular and atomic level.
8. _____ chemistry is important for the production of materials like plastics, fertilizers, and medicines.



Tick(✓) for True and cross (x) for False statements

1. Chemistry is only concerned with the study of living organisms.

☐

2. Jabir ibn Hayyan contributed to the development of laboratory techniques.

☐

3. Physical chemistry studies how matter behaves at the molecular and atomic level.

☐

4. Organic chemistry deals with substances that do not contain carbon.

☐

5. Industrial chemistry is important for large-scale production of chemicals.

☐

6. The study of enzymes is part of biochemistry.

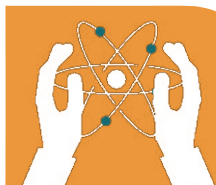
☐

7. Robert Boyle is known for developing the periodic table.

☐

8. Inorganic chemistry deals with substances like metals, salts, and minerals.

☐



Choose the correct answer.

1. Who is known as the "father of modern chemistry"?

- a) Jabir ibn Hayyan b) Robert Boyle
- c) Marie Curie d) Aristotle

2. Which branch of chemistry studies substances that contain carbon?

- a) Inorganic Chemistry b) Organic Chemistry
- c) Physical Chemistry d) Biochemistry

3. What is the main focus of industrial chemistry?

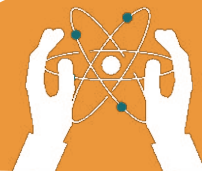
- a) Chemical reactions in organisms
- b) The study of life processes
- c) Large-scale production of chemicals
- d) The properties of gases

4. Who discovered radium and polonium?

- a) Robert Boyle b) Aristotle
- c) Marie Curie d) Plato

5. What does biochemistry study?

- a) Chemical reactions in living organisms
- b) The properties of metals
- c) The behaviour of gases
- d) The structure of organic compounds



6. Which branch of chemistry studies the relationship between pressure and volume of gases?

- a) Organic Chemistry b) Physical Chemistry
- c) Biochemistry d) Inorganic Chemistry

7. What did Robert Boyle contribute to chemistry?

- a) Developed the atomic theory
- b) Discovered the periodic table
- c) Boyle's Law, the relationship between pressure and volume of gas
- d) Created the first chemical formula

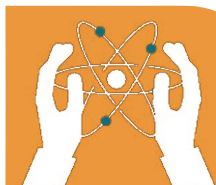
8. Which of these is not an example of an organic compound?

- a) Glucose b) Water
- c) Protein d) DNA

Answer the Following Questions:

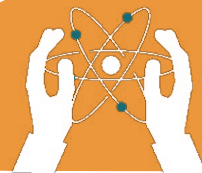
Short-Answer Questions.

- 1. Name two early scientists who contributed to the field of chemistry.**
- 2. What is chemistry?**
- 3. Write the difference between organic and inorganic chemistry?**
- 4. What is the importance of chemistry in daily life?**



Long-Answer Questions.

- 1. Describe the branches of chemistry.**
- 2. Discuss the contributions of Robert Boyle and Marie curie to modern chemistry.**
- 3. Describe the different branches of chemistry and their significance in real life.**



Hands-on or Project-Based



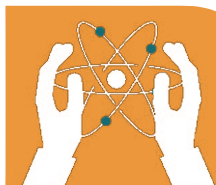
ACTIVITY

1. Create a simple project where students demonstrate the process of distillation (e.g., distilling water) to show how early scientists like Jabir ibn Hayyan helped develop chemical processes.
2. Students can make a model of a molecule of a common organic compound (e.g., methane, glucose) using sticks and balls or marshmallows and toothpicks.



Critical Thinking

How do you think the contributions of early scientists like Jabir ibn Hayyan influenced modern chemistry and industries today? Give examples of how chemistry affects our daily lives.



FUNDAMENTALS OF CHEMISTRY

How much do you know about chemistry? Complete the statements below with the correct word from the pool of words inside the Erlenmeyer flask on the next page.

1. The practice of _____, which involves the separation of metals from their ores, saw significant development.
2. _____ chemistry is employed in the manufacture of a wide range of products, including paint, fertilizer, and sun block.
3. The _____ is a subatomic particle that carries an electrical charge of positive one.
4. Chemists who work in _____ examine the non-biological trace evidence that might be found at crime scenes to determine the identity of unknown compounds.
5. The third constituent of an atom is called a _____, and it has a charge of negative one.
6. Chemistry is the study of _____, which can be described as anything that has mass and takes up space.
7. Lavoisier developed an experimentally based theory of the chemical reactivity of _____ and co-authored the modern system for naming chemical substances.

